

Peter Nguyen

Senior Software Engineer

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Summary

Senior Software Engineer with 8 years of experience building backend systems for enterprise workflows, healthcare digital platforms, and identity-driven transformation programs. Strong expertise in Ruby on Rails, PostgreSQL, Redis, Sidekiq, and distributed backend architectures supporting high-volume user interactions. Experienced in integrating Python-based machine learning services using PyTorch for intent classification, personalization, and conversational AI pipelines. Focused on building user-centric, API-driven systems with strong ownership mindset and delivery in fast-paced distributed engineering environments.

Skills

- **Backend Engineering:** Ruby on Rails, REST APIs, GraphQL, MVC architecture
- **Databases:** PostgreSQL, Redis, data modeling, indexing, query optimization
- **Async & Processing:** Sidekiq, background jobs, message queues
- **AI / ML Integration:** Python, PyTorch, Stable Diffusion, NLP pipelines
- **System Architecture:** API design, event-driven architecture, session management, microservice integration
- **DevOps & Infrastructure:** Docker, AWS, CI/CD pipelines, Linux environment
- **Engineering Tools:** Git, RSpec, monitoring, logging systems
- **Collaboration:** Agile delivery, ownership mindset, cross-functional communication, product thinking

Work Experience

Accenture • Senior Software Engineer • Aug 2024 - Present

- Designed backend architecture for “**LumenEcho Conversational Experience Engine**” enabling AI-driven conversational workflows using **Ruby on Rails** and RESTful API services.
- Built session persistence and conversation memory layer using **PostgreSQL**, supporting around 12 million stored conversational records across distributed environments.
- Implemented asynchronous message processing pipelines using **Sidekiq** and **Redis** to handle queue-based AI response generation and background inference tasks.
- Integrated Python-based ML microservices using **PyTorch** for intent classification and response ranking in conversational flows.
- Developed multimodal AI feature hooks integrating **Stable Diffusion** for dynamic content generation within user conversations.
- Optimized API orchestration layer for handling approximately 3 million daily conversation requests across distributed services.
- Collaborated on secure and moderated response processing layer ensuring controlled AI output behavior aligned with product interaction rules.

Novo Nordisk • Senior Software Engineer • Apr 2021 - Aug 2024

- Developed patient engagement backend services for chronic disease management platforms using **Ruby on Rails** and **PostgreSQL**.
- Built structured treatment tracking systems supporting around 1.5 million patient interaction records for diabetes and obesity care journeys.
- Implemented background processing workflows using **Sidekiq** for medication reminders, onboarding flows, and adherence tracking.
- Designed secure API services integrating healthcare content delivery systems and patient education modules.
- Improved Redis-based caching strategies for patient session state and engagement tracking data across distributed services.
- Supported HCP-facing digital portal features for chronic disease education and treatment guidance workflows.
- Contributed to system reliability improvements for long-running patient engagement processes across multiple regional deployments.

Deloitte • Application Engineer • Dec 2018 - Mar 2021

- Developed enterprise identity and access workflow services using **Ruby on Rails** for role-based access request and approval systems.
- Built identity lifecycle management components supporting around 500,000 enterprise user accounts across integrated systems.
- Implemented backend services for access approval routing, password reset flows, and identity verification processes.
- Developed transformation program dashboards for tracking delivery milestones, dependencies, and operational risks across enterprise initiatives.
- Designed Redis-based session handling mechanisms for authentication workflows and access request pipelines.
- Integrated backend services with enterprise audit logging systems for compliance tracking and operational reporting.

Objectivity Inc • Web Developer • Jul 2017 - Nov 2018

- Built backend workflow services for financial operations modernization platforms using **Ruby on Rails** and relational databases.
- Developed case management and operational workflow modules supporting around 200,000 financial service records.
- Implemented structured data access layers improving legacy system integration across financial workflows.
- Supported internal tooling features for enterprise workplace portal applications used for operational service requests.
- Optimized database queries and indexing strategies in **PostgreSQL** for workflow-heavy transaction systems.
- Contributed to UI-driven backend APIs for workplace experience portals supporting internal enterprise users.

Education

- **Hanoi University of Science and Technology • Bachelor's Degree • Computer Programming • Hanoi, Vietnam • 2013 - 2017**